

B. Tech Degree IV Semester Examination, April 2010**CS/EB/EC/EI 402 MICROPROCESSORS**
(2006 Scheme)

Time: 3 Hours

Maximum Marks: 100

PART A(Answer ALL questions)

(8 x 5 = 40)

- I. a. Write a short note on usage of temporary registers in 8085.
 b. Write the usages of 8085 pins (i) READY (ii) ALE
 c. Differentiate between the op-codes LHLD and LDAX.
 d. Write a short note on various rotate instructions and their functionalities in 8085.
 e. Write a short note on (i) T State (ii) Machine cycle
 f. Draw an Instruction Fetch Timing Diagram of 8085.
 g. Write a short note on various ports available in 8255.
 h. Write a note on Transmitter section of 8251 peripheral device.

PART B

(4 x 15 = 60)

- II. a. Differentiate between I/O Mapped I/O and Memory mapped I/O mapping techniques. (8)
 b. Describe the Register organization of 8085 in detail. (7)
- OR**
- III. Write the functions of following group of pins in 8085. (15)
 (i) S0, S1 (ii) HOLD, HLDA (iii) SID, SOD
- IV. a. Explain various addressing modes available in 8085 with the help of examples. (8)
 b. Design and write a Delay routine for 1 second delay. Assume the clock speed of microprocessor is 3 MHz. (7)
- OR**
- V. a. Write an assembly Language program in 8085 to arrange a set of numbers stored from memory address 3001 onwards in ascending order and number of elements in the set is available in location address 3000. (10)
 b. Write a short note on stack based op-codes and how will it modify the contents of register SP. (5)
- VI. Draw and explain the timing diagram for the instruction LDA 3000. (15)
- OR**
- VII. Describe various types of interrupts and their organization in 8085. (15)
- VIII. a. Write the concept of memory interfacing with the help of an example. (8)
 b. Draw the architecture and register organization of peripheral chip 8251. (7)
- OR**
- IX. a. Describe various operating modes and control words of peripheral chip 8255. (5)
 b. Design an interface circuit to accept 8 bits of data from a ADC and produce various control signals according to
- (i) the data read is less than 128
 (ii) The data read is equal to 128
 (iii) The data read is greater than 128

Also write the program for implementing the interfacing circuit.

